



# Blockchain Principles and Applications

Amir Mahdi Sadeghzadeh, PhD

Data and Network Security Lab (DNSL)  
Trustworthy and Secure AI Lab (TSAIL)

# Recap

# Proof of Work

Find **nonce** such that

$H(\text{hash}(\text{parent.header}), \text{Merkle root of tx}, \text{nonce}) < \text{threshold}$

**Finding** entails **work**

**Nonce** is the **proof**

Doing this work allows miners to **participate meaningfully** in the protocol

PoW is a Sybil and spam resistant leader election mechanism

# Proof of Stake

Allow meaningful participation based on stake  
block proposers own coins

Level of participation proportional to stake  
higher probability of being a proposer

Doing work is replaced by owning coins  
energy efficient  
capital efficient – no need for mining hardware

# Idea 1

PoS attempt 1

$H(\text{hash}(\text{parent.header}), \text{Merkle root of tx}, \text{public key}) < \text{threshold} \times \text{stake}$

# Idea 1

## PoS attempt 1

$H(\text{hash}(\text{parent.header}), \text{Merkle root of tx}, \text{public key}) < \text{threshold} \times \text{stake}$

## Problem: Grinding

can try different set of tx such that Merkle root of tx works out “correctly”

so probability is not purely proportional to stake

# Idea 2

## PoS attempt 2

$H(\text{hash}(\text{parent.header}), \text{public key}) < \text{threshold} \times \text{stake}$

Got rid of transaction hash

## Problem: Liveness

just one trial to form a block

unlike PoW where nonce can be tried at will

# Idea 3

PoS attempt 3

$H(\text{hash}(\text{parent.header}), \text{timestamp}, \text{public key}) < \text{threshold} \times \text{stake}$



# Idea 3

## PoS attempt 3

$H(\text{hash}(\text{parent.header}), \text{timestamp}, \text{public key}) < \text{threshold} \times \text{stake}$

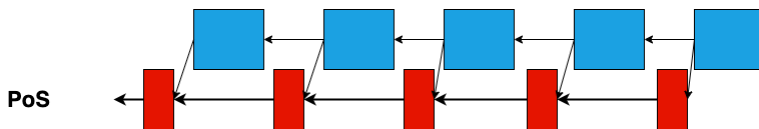
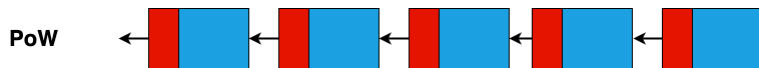
## Problems:

1. Block content is not tamper-resistant against an adaptive adversary
2. Public election: vulnerable to bribery/corruption, not resistant to adaptive adversary

# Idea 3

## PoS attempt 3

$H(\text{hash}(\text{parent.header}), \text{timestamp}, \text{public key}) < \text{threshold} \times \text{stake}$



Block header



Block content

# Proof of Stake

# Idea 4

To solve the predictability of the winner in the future:

PoS attempt 4

$\text{hash}(\text{hash}(\text{parent.header}), \text{timestamp}, \text{secret key}) < \text{threshold} \times \text{stake}$

What is the problem?

# Verifiable Random Function (VRF)

Verifiable Random Function (VRF)

$$\text{VRF}( \text{sk} , x ) \rightarrow ( y , \pi )$$

$$\text{Verify}( \text{pk} , x , y , \pi ) \rightarrow \text{True/False}$$

- used for signing blocks
- helps achieve **adaptive security**

attacker corrupts upcoming blocks well in advance

# Idea 4

PoS attempt 4

$\text{VRF}(\text{hash}(\text{parent.header}), \text{timestamp}, \text{secret key}) < \text{threshold} \times \text{stake}$

# Attack: Nothing at Stake

- In PoS, there's **little to no cost** to sign blocks on all forks.
- Rational validators (miners in PoW) might sign blocks on **all possible forks** to **maximize their rewards**, since there's no significant downside.
- **The Attack:**
  - This undermines **finality** and **security** because validators aren't incentivized to converge on a single chain.
  - Attackers can exploit this by intentionally creating forks and encouraging validators to sign all of them.
  - This can lead to **double-spending attacks**, where the same coins are spent in different forks.

# Attack: Nothing at Stake

$\text{VRF}(\text{hash}(\text{parent.header}), \text{timestamp}, \text{secret key}) < \text{threshold} \times \text{stake}$

Longest chain rule

parent is tip of longest chain

Adversary deviates

can grow on all blocks (even Genesis)

No computation limit to deviation

unlike PoW

Nothing at Stake (NaS)



# NaS Tree

Honest participants

grow chain as a Poisson process

growth rate linear in time

Adversary

grows a tree of blocks

number of blocks grows exponentially in time

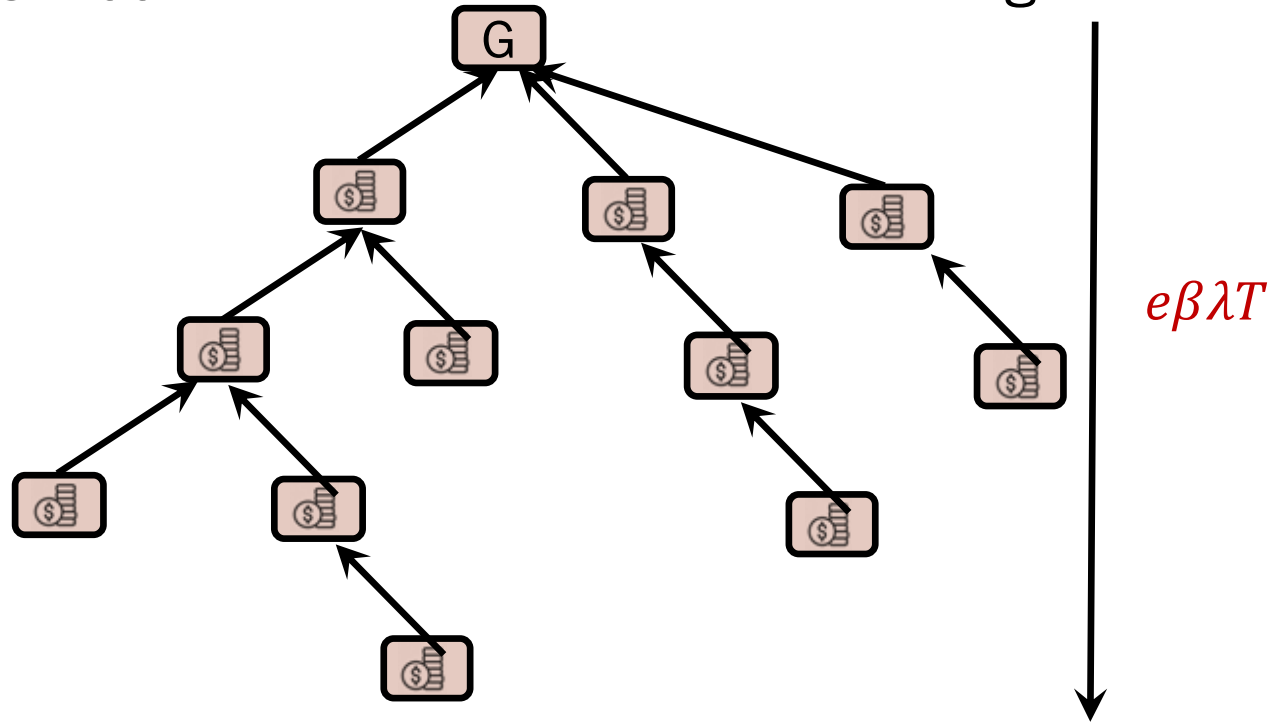
NaS

allows adversary to compete with honest participants unevenly

# NaS Tree and Longest Chain

NaS Tree

Longest chain



# Security of PoS Longest Chain

Honest participants

grow chain as a Poisson process

growth rate linear in time  $(1 - \beta)\lambda T$

Adversary

grows a NaS tree in private

longest chain length  $e\beta\lambda T$

Security against Private attack

$$(1 - \beta)\lambda T > e\beta\lambda T \quad \text{or} \quad \beta < \frac{1}{1+e} \approx 27\%$$

# Security

## Security against Private attack

$$\beta < \frac{1}{1+e} \approx 27\%$$

## Secure against all attacks

 Cornell University

arXiv.org > cs > arXiv:1910.02218

Search...  
Help | Advan

Computer Science > Cryptography and Security

**Proof-of-Stake Longest Chain Protocols: Security vs Predictability**

Vivek Bagaria, Amir Dembo, Sreeram Kannan, Sewoong Oh, David Tse, Pramod Viswanath, Xuechao Wang, Ofer Zeitouni

*(Submitted on 5 Oct 2019 (v1), last revised 23 Feb 2020 (this version, v3))*

The Nakamoto longest chain protocol is remarkably simple and has been proven to provide security against any adversary with less than 50% of the total hashing power. Proof-of-stake (PoS) protocols are an energy efficient alternative; however existing protocols adopting Nakamoto's longest chain design achieve provable security only by allowing long-term predictability (which have serious security implications). In this paper, we prove that a natural longest chain PoS protocol with similar predictability as Nakamoto's PoW protocol can achieve security against any adversary with less than  $1/(1+e)$  fraction of the total stake. Moreover we propose a new family of longest chain PoS protocols that achieve security against a 50% adversary, while only requiring short-term predictability. Our proofs present a new approach to analyzing the formal security of blockchains, based on a notion of adversary-proof convergence.

Comments: 65 pages, 16 figures

Subjects: **Cryptography and Security (cs.CR)**

Cite as: arXiv:1910.02218 [cs.CR]  
(or arXiv:1910.02218v3 [cs.CR] for this version)

**Bibliographic data**  
[Enable Bibex (What is Bibex?)]

**Submission history**  
From: Xuechao Wang [view email]

# Boosting Security Threshold

Security against all attacks

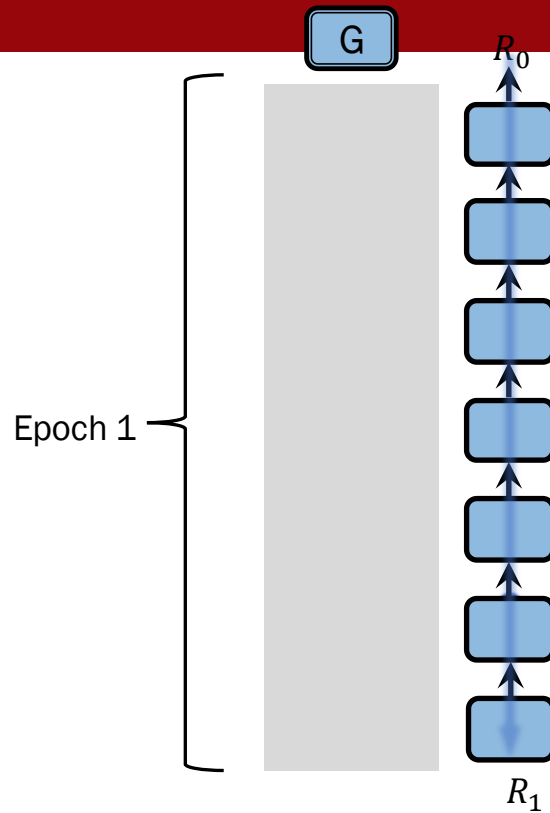
$$\beta < \frac{1}{1+e} \approx 27\%$$

Key idea: Reduce the number of randomness sources

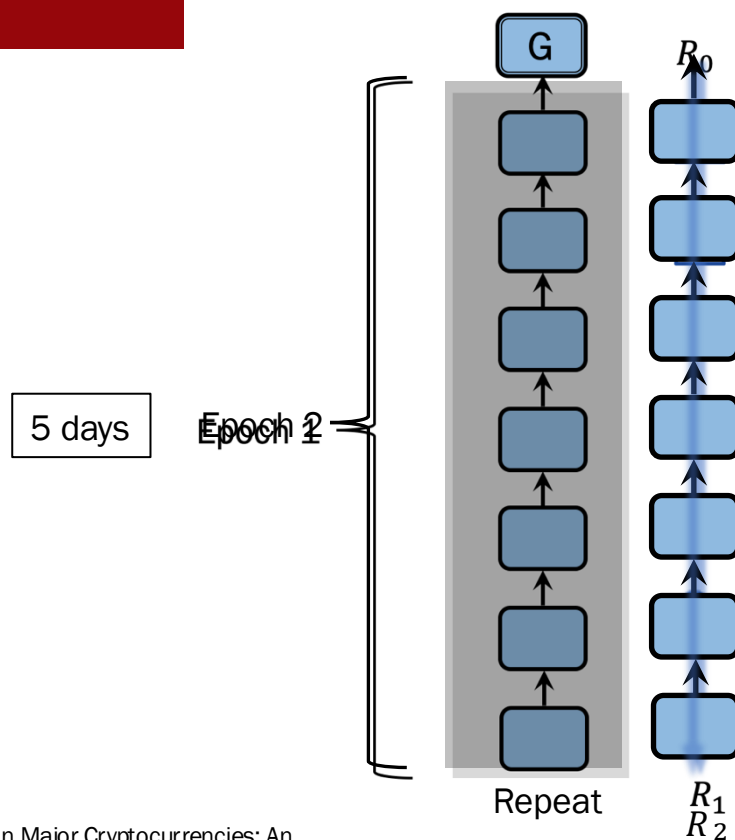
Idea 1: Only use randomness from genesis block

VRF(hash(Genesis), timestamp, secret key) < threshold x **stake**

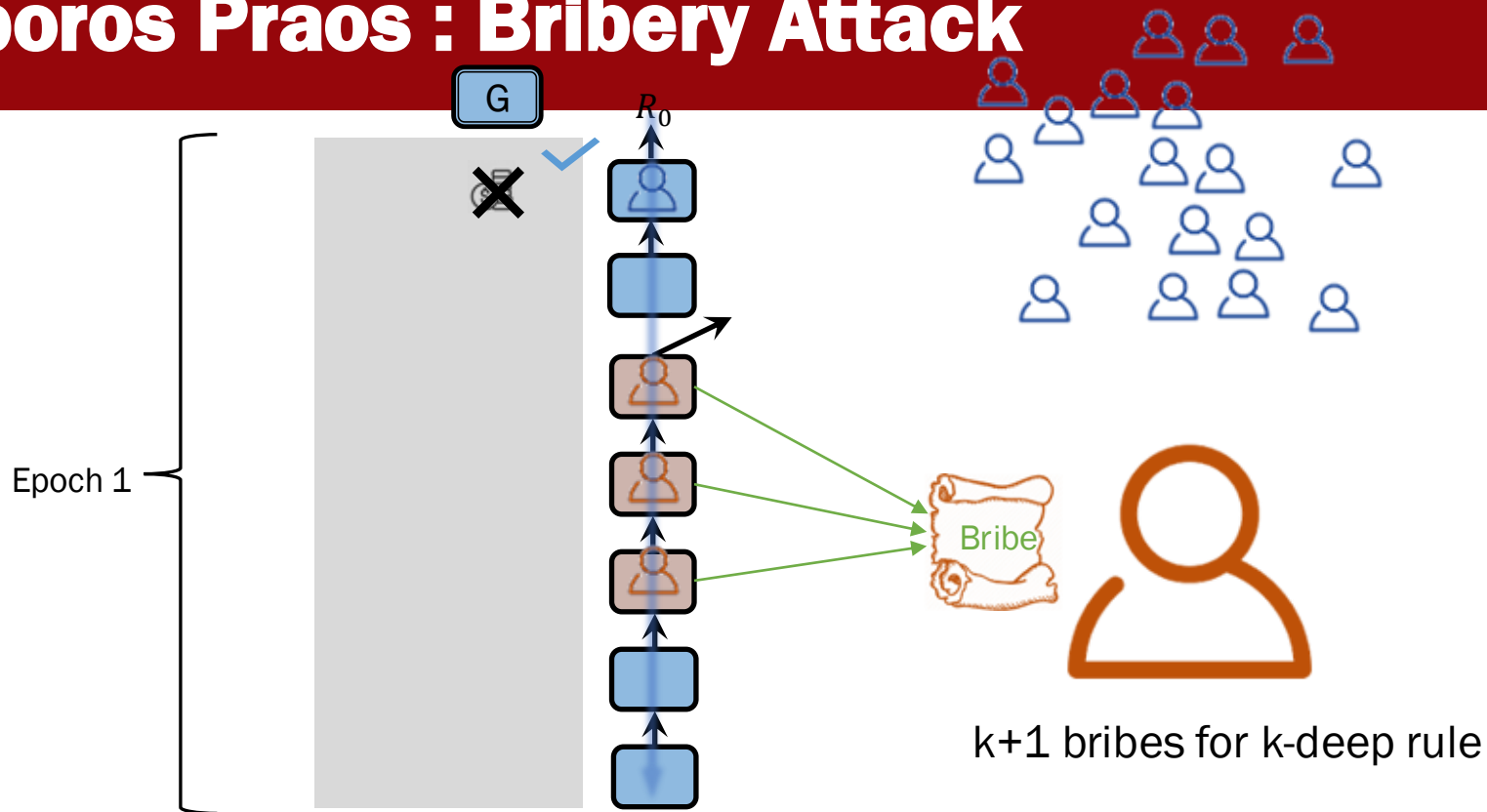
# Ouroboros Praos



# Ouroboros Praos



# Ouroboros Praos : Bribery Attack





# Boosting Security Threshold

Security against all attacks

$$\beta < \frac{1}{1+e} \approx 27\%$$

Key idea: Reduce the number of randomness sources

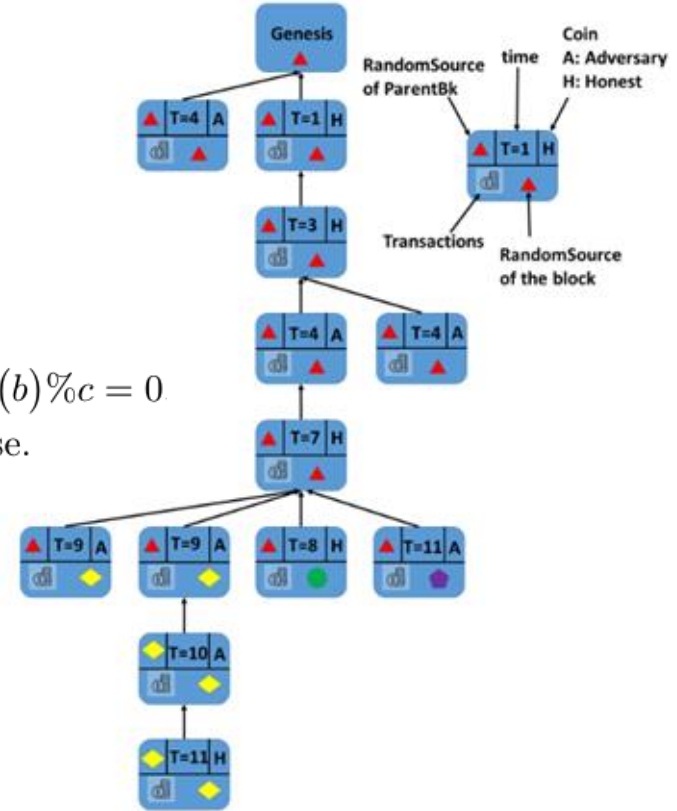
Idea 2: c-correlation

# c-correlation

- Update the randomness of a block only at blocks with height multiples of  $c$

$$\text{RandSource}(b) := \begin{cases} \text{VRF}(\text{RandSource}(\text{parent}(b)), ts, sk), & \text{if } \text{depth}(b) \% c = 0 \\ \text{RandSource}(\text{parent}(b)), & \text{otherwise.} \end{cases}$$

$$\text{VRF}(\text{RandSource}(\text{parent}), ts, sk_n) < T \cdot \text{stake}_n$$

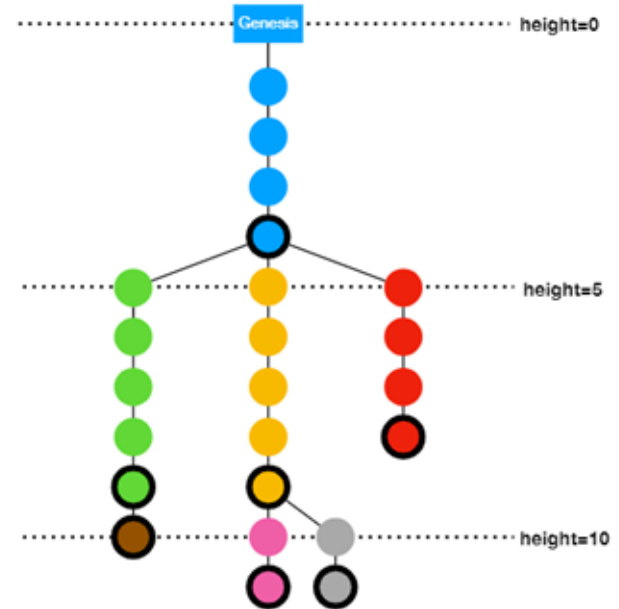


# Analysis of private NaS

- Godfather-block:  $\text{height}(b) \% c = 0$
- Only fork at the parents of godfather-blocks

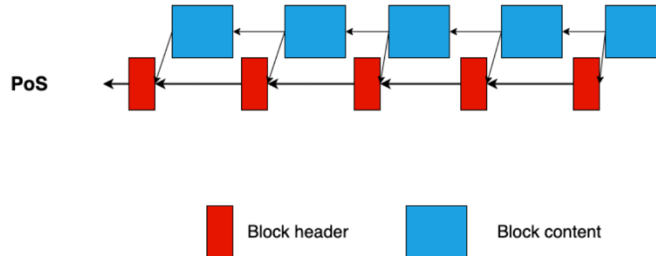
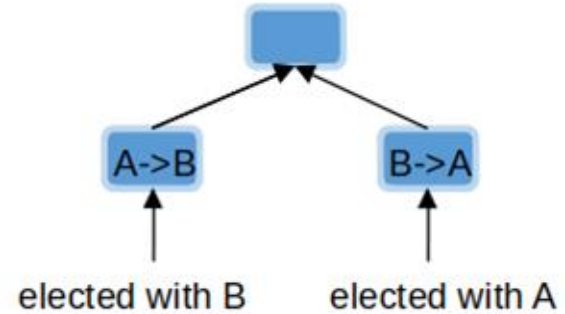
| $c$       | 1               | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      |
|-----------|-----------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| $\phi_c$  | e               | 2.22547 | 2.01030 | 1.88255 | 1.79545 | 1.73110 | 1.68103 | 1.64060 | 1.60705 | 1.57860 |
| $\beta_c$ | $\frac{1}{1+c}$ | 0.31003 | 0.33219 | 0.34691 | 0.35772 | 0.36615 | 0.37299 | 0.37870 | 0.38358 | 0.38780 |

$\beta_c = 1/(1+\phi_c)$ : threshold for private NaS attack



# Dynamic stake

- Flaw of static stake
  - Prevent nodes from leaving and joining
  - A coin with no actual stake can participate
- What if stake is updated immediately?
  - Grinding attack: once the adversary is elected as a leader at round  $i$ , it can include transactions at round  $i$  to transfer all stake to a coin that has a higher chance of winning the election at round  $i + 1$
- What if stake is updated with a delay of  $s$  blocks?
  - Long range attack: have a private chain with  $s$  blocks



# Resources

- ECE/COS 470, Pramod Viswanath, Princeton 2024